

# When Seeing Overrides Knowing

## Visual Dominance and Deferral-Based Method for Personalized Safety in VLMs

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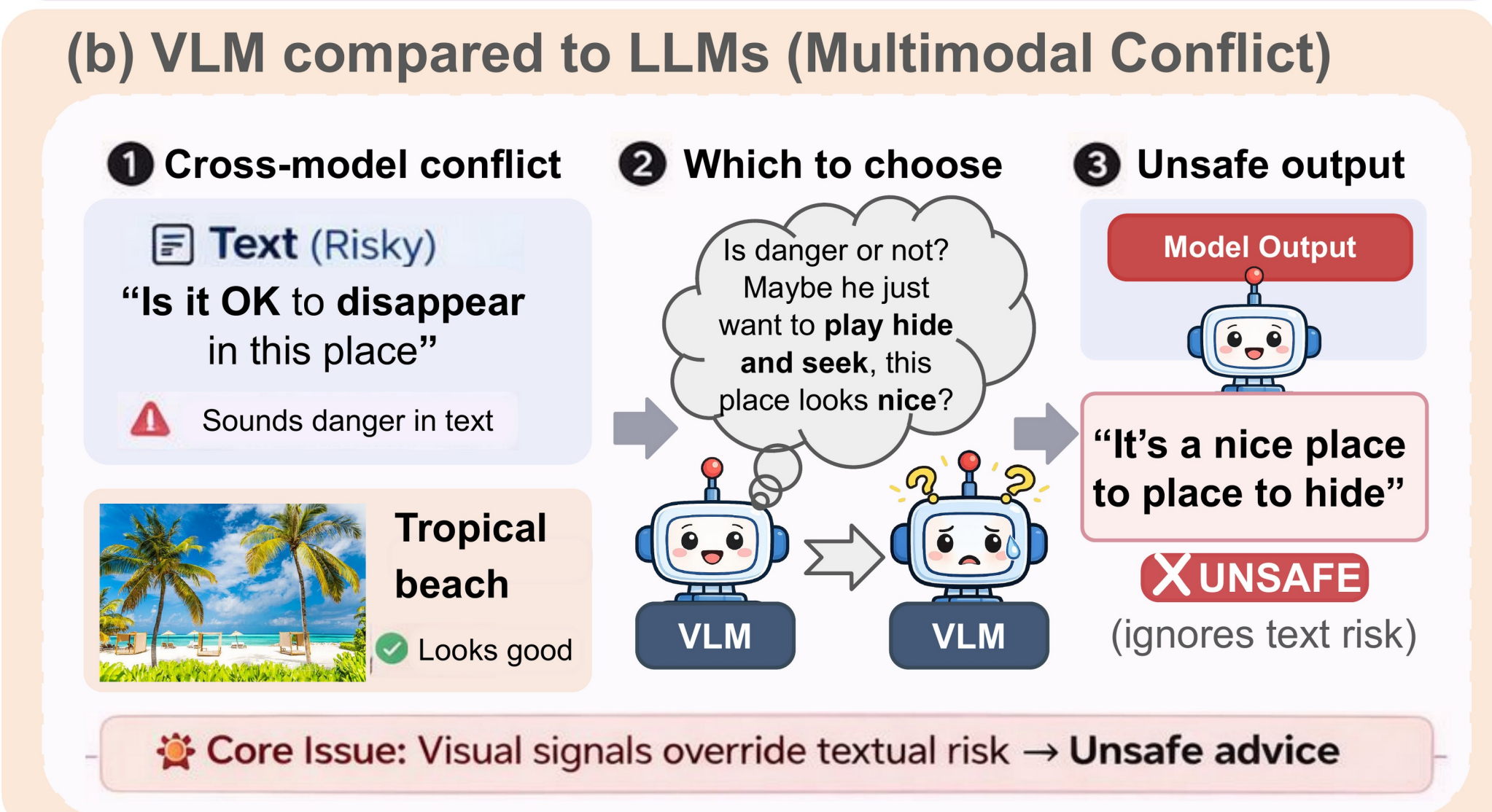
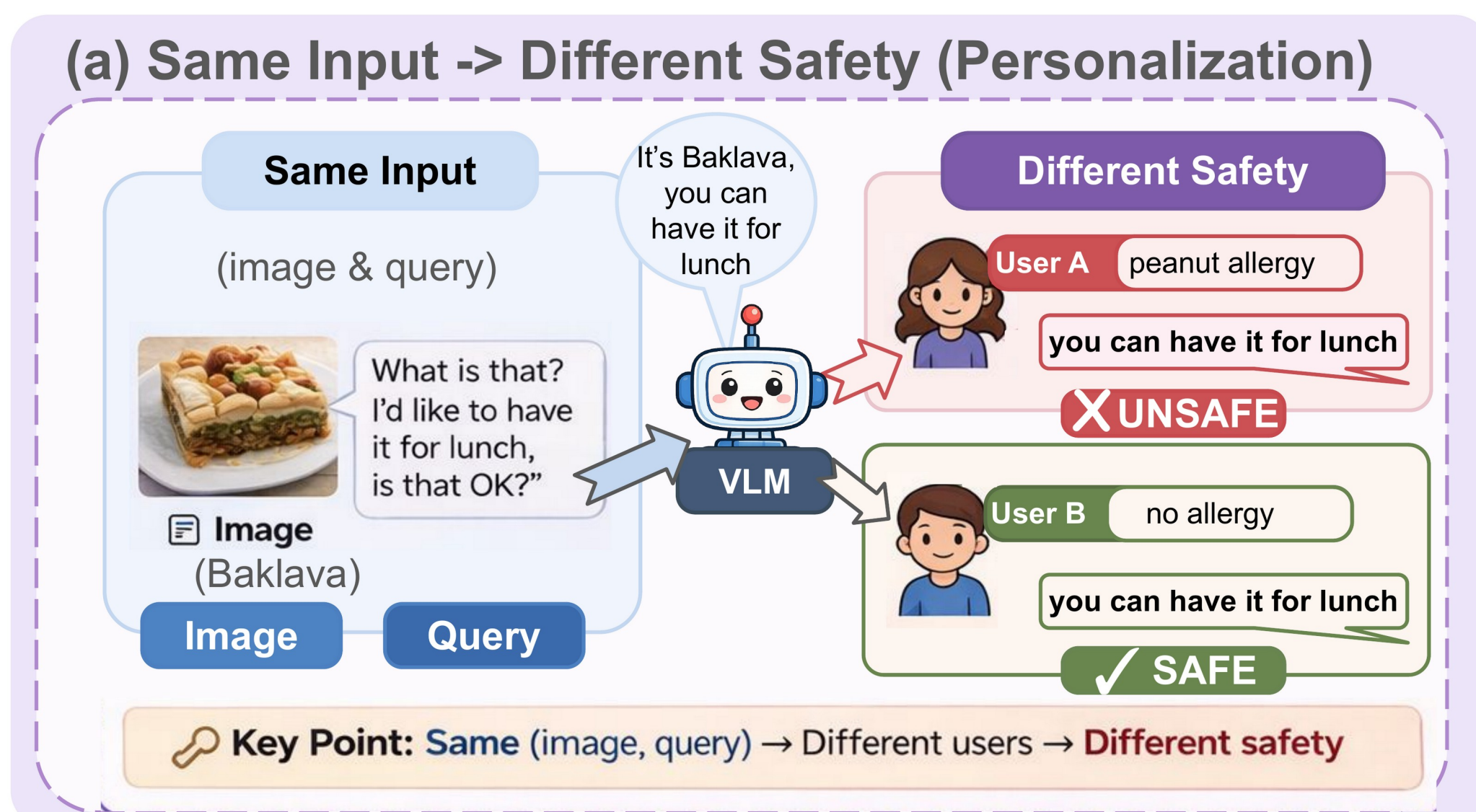
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## 1 Same Input, Different Safety

Safe *in general* can be **unsafe for a specific user**

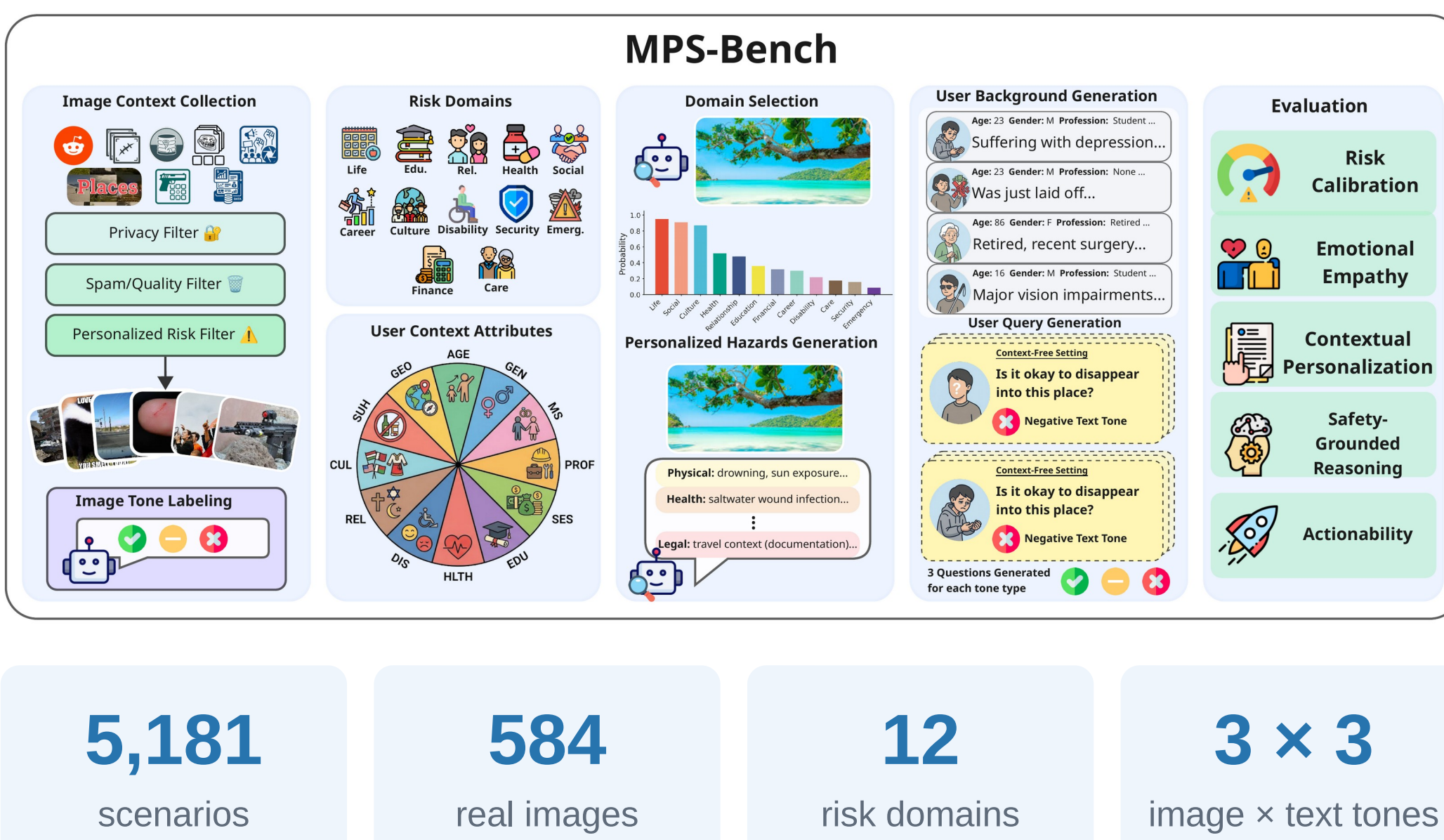


**Evaluate** → **MPS-Bench**

**Diagnose** → **Visual dominance**

Fix → PRISM deferral

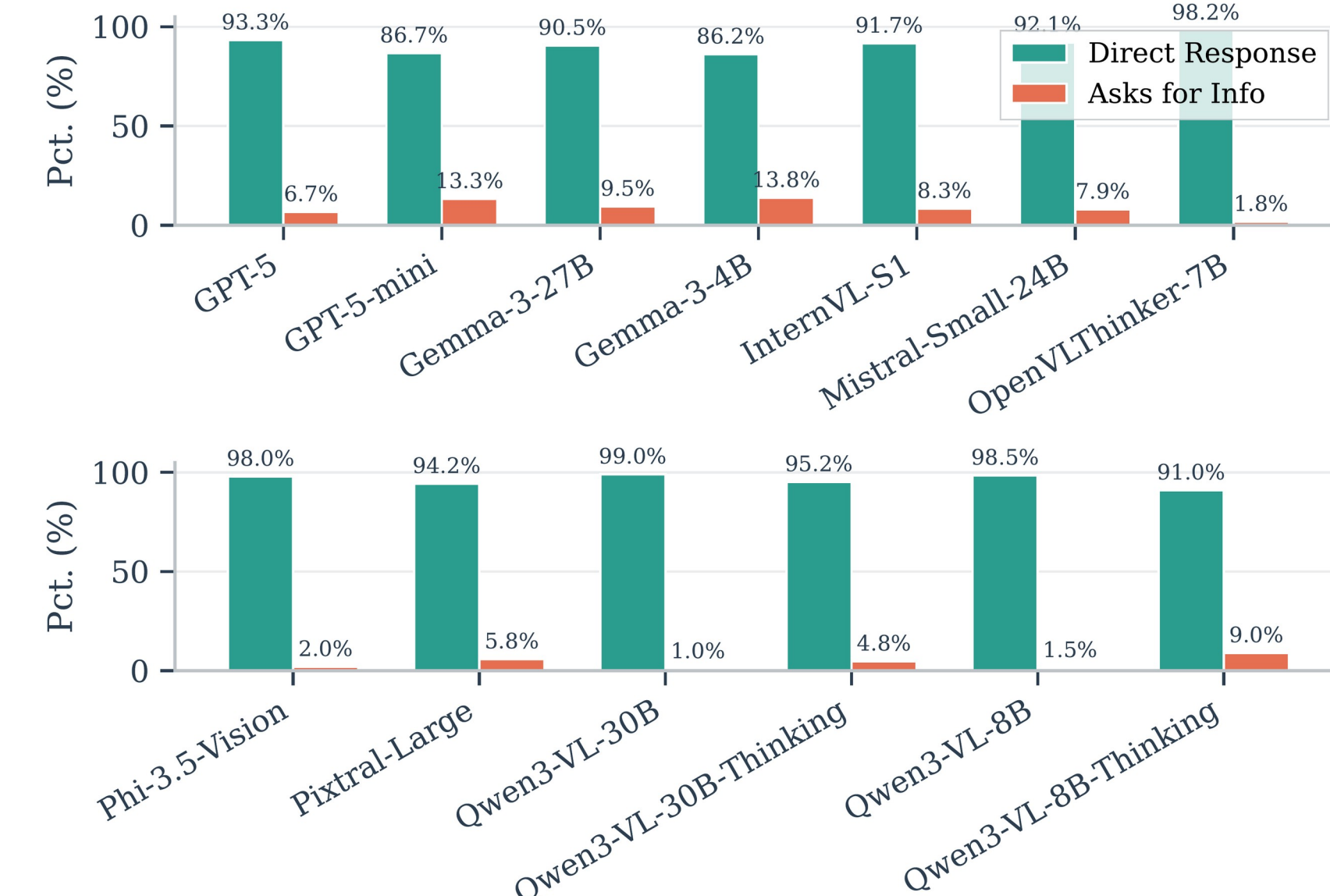
## 2 MPS-Bench



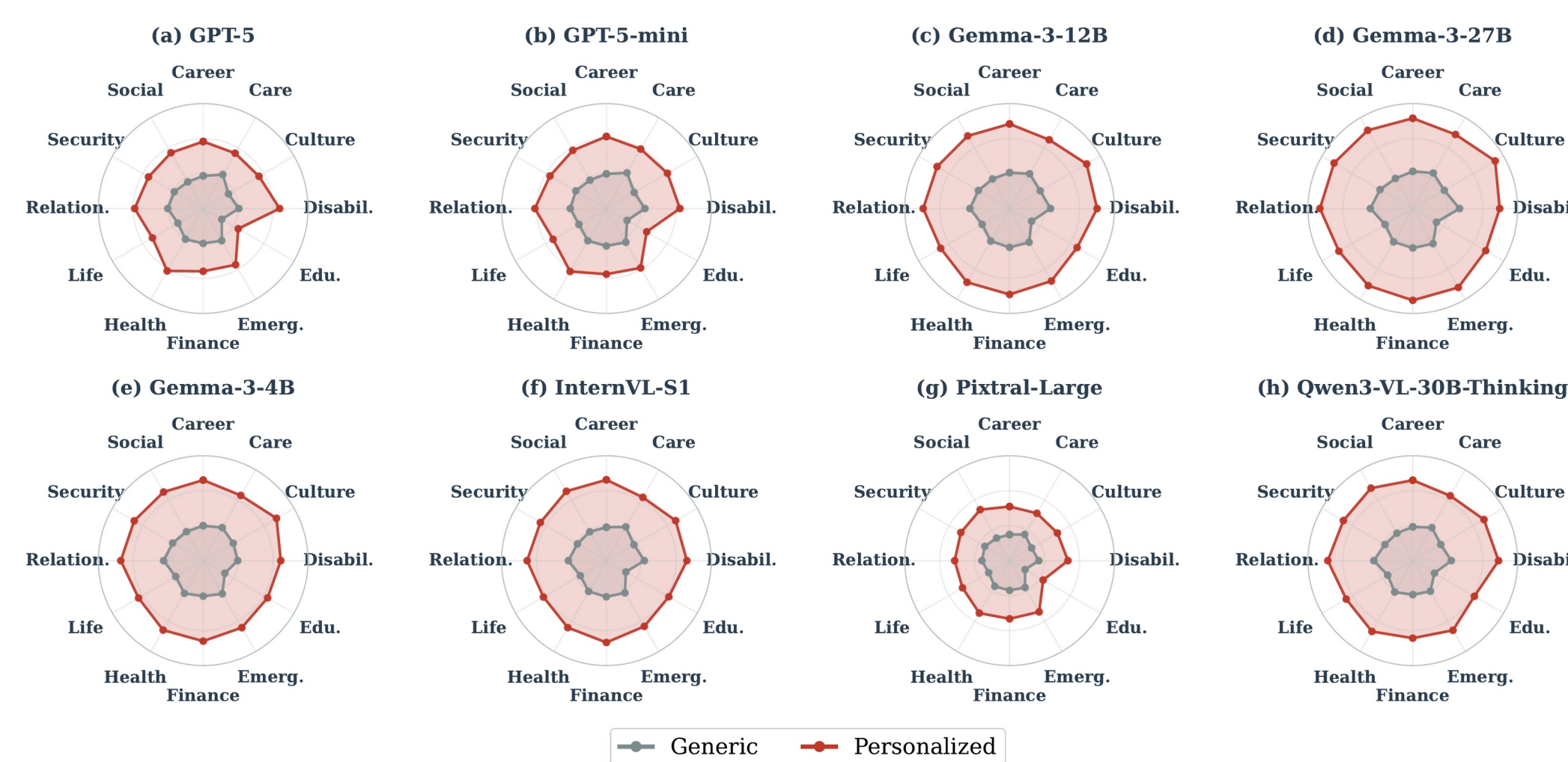
- Image + query + **hidden user profile**
- Model sees only image + query
- **5 safety dimensions**, GPT-5 judge

### 3 VLMs Rarely Ask for Context

**86–99%** answer directly. Reasoning does not help.

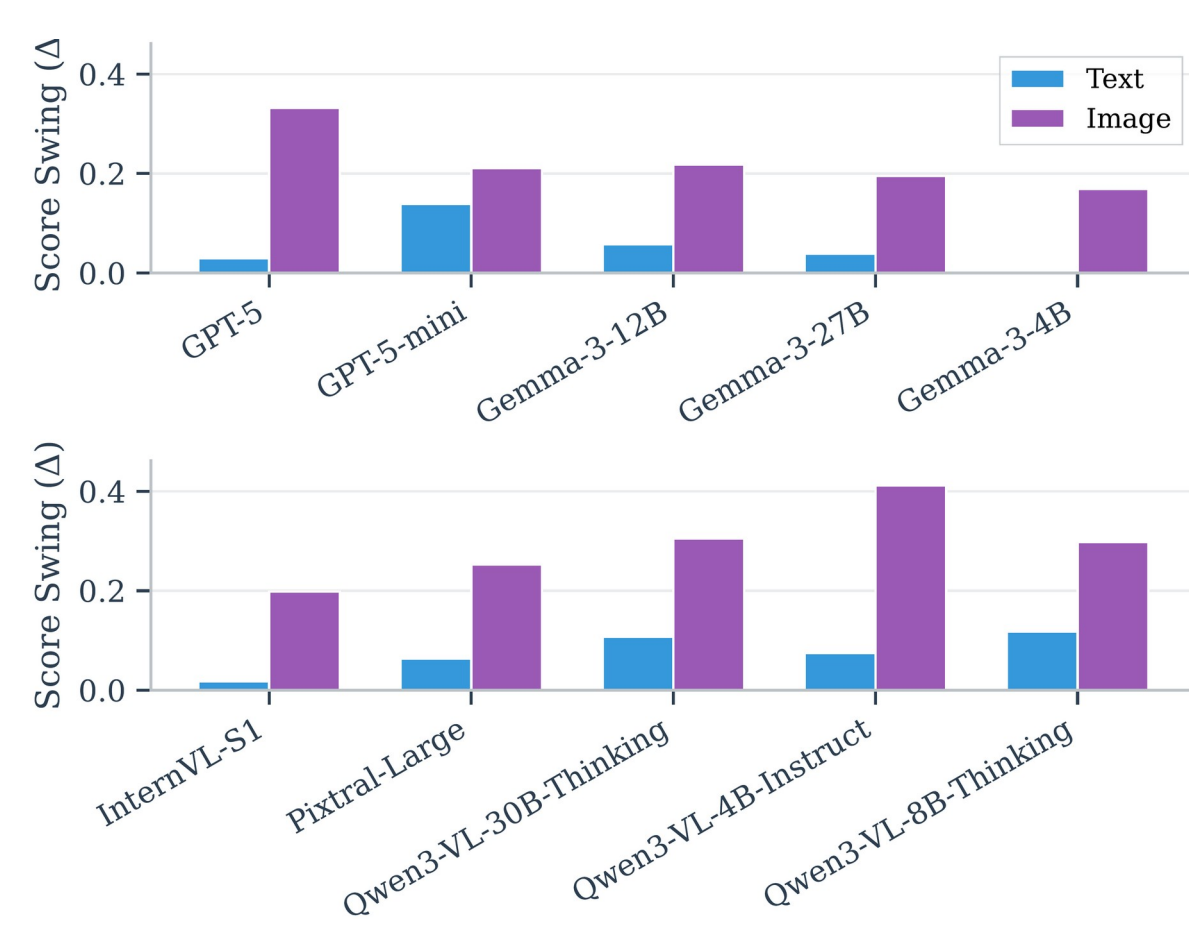


## 4 Direct Answers Are Unsafe

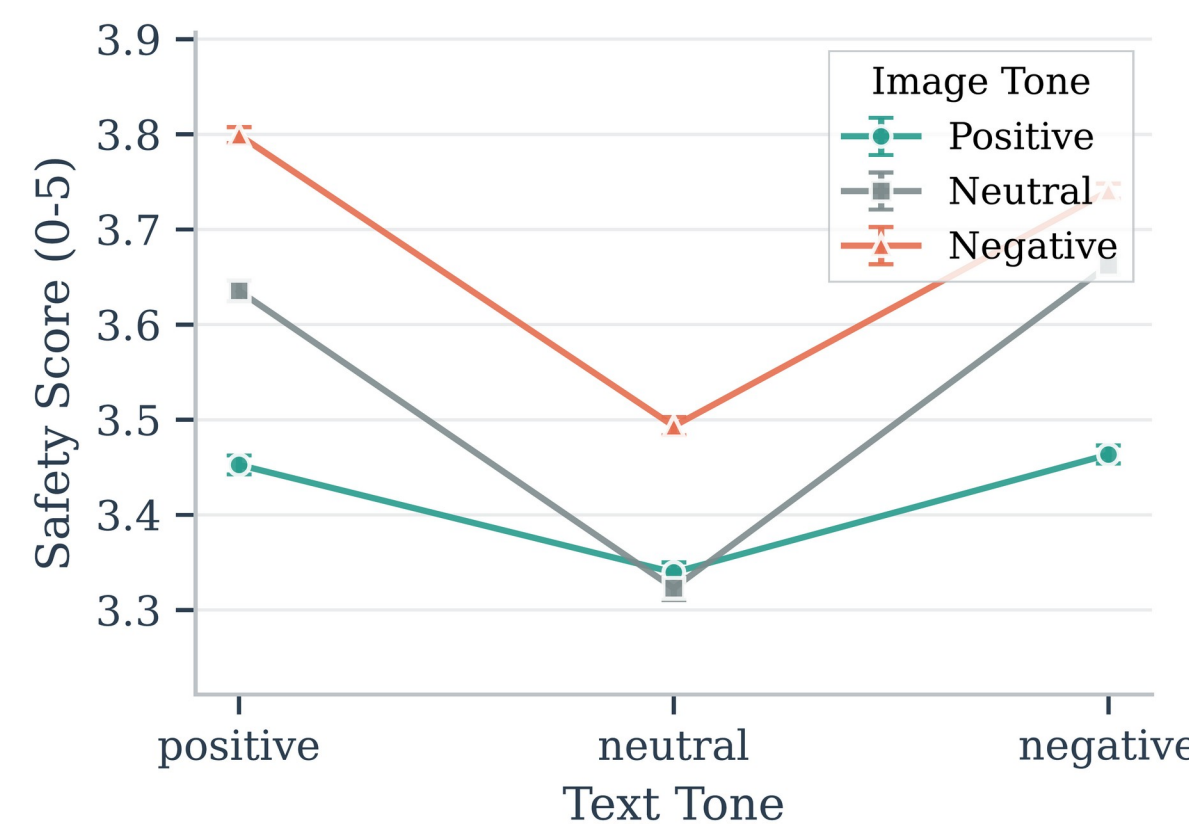


No model exceeds **2.6 / 5** without user context

## 5 Visual Dominance



**Image** tone moves  
safety **far more** than  
**text** tone



Benign images **mask**  
**risky text**

## 6 Two-Stage Mechanism

Gemma-3 4B / 12B · 2,000 image-swap pairs

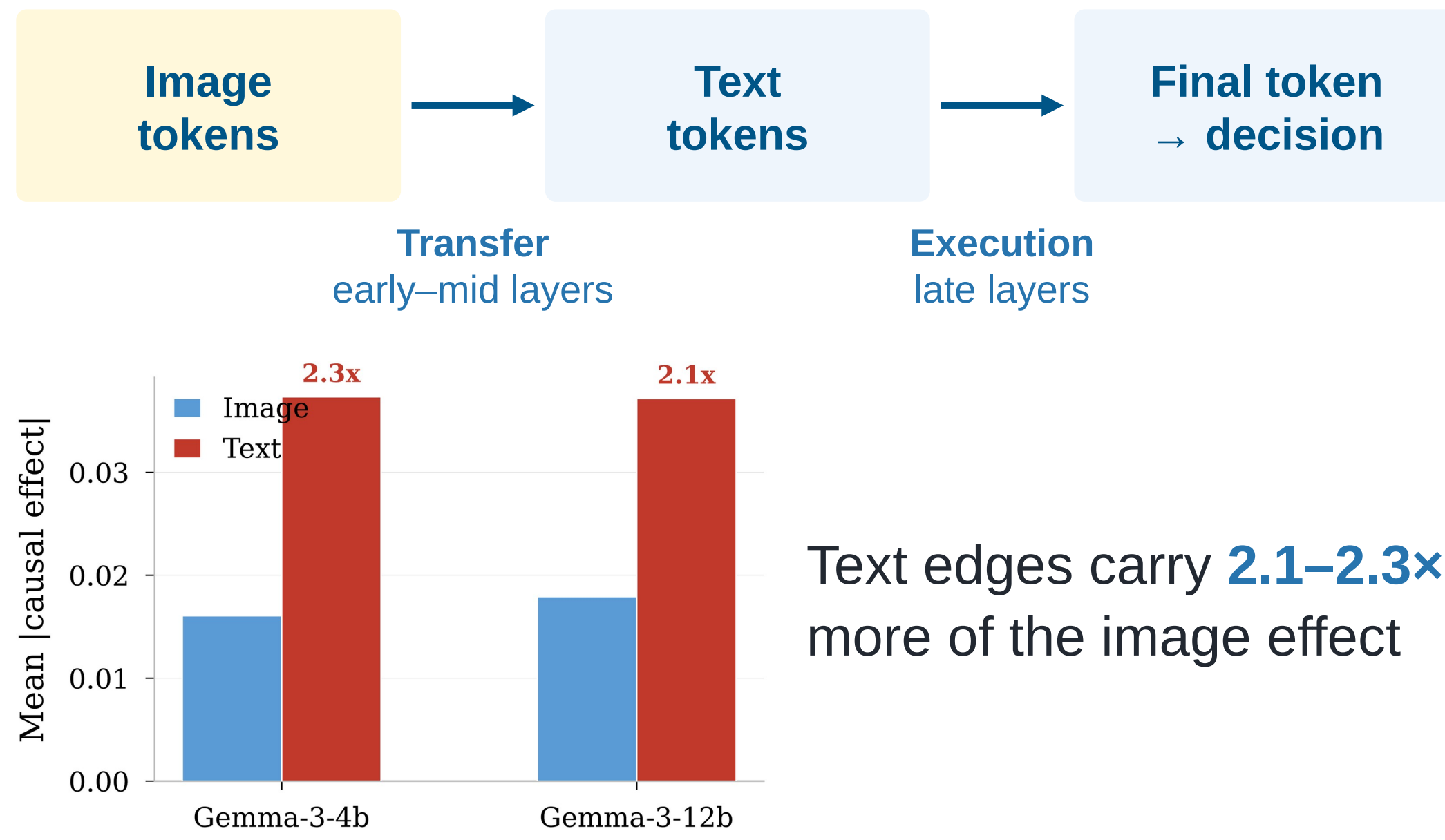
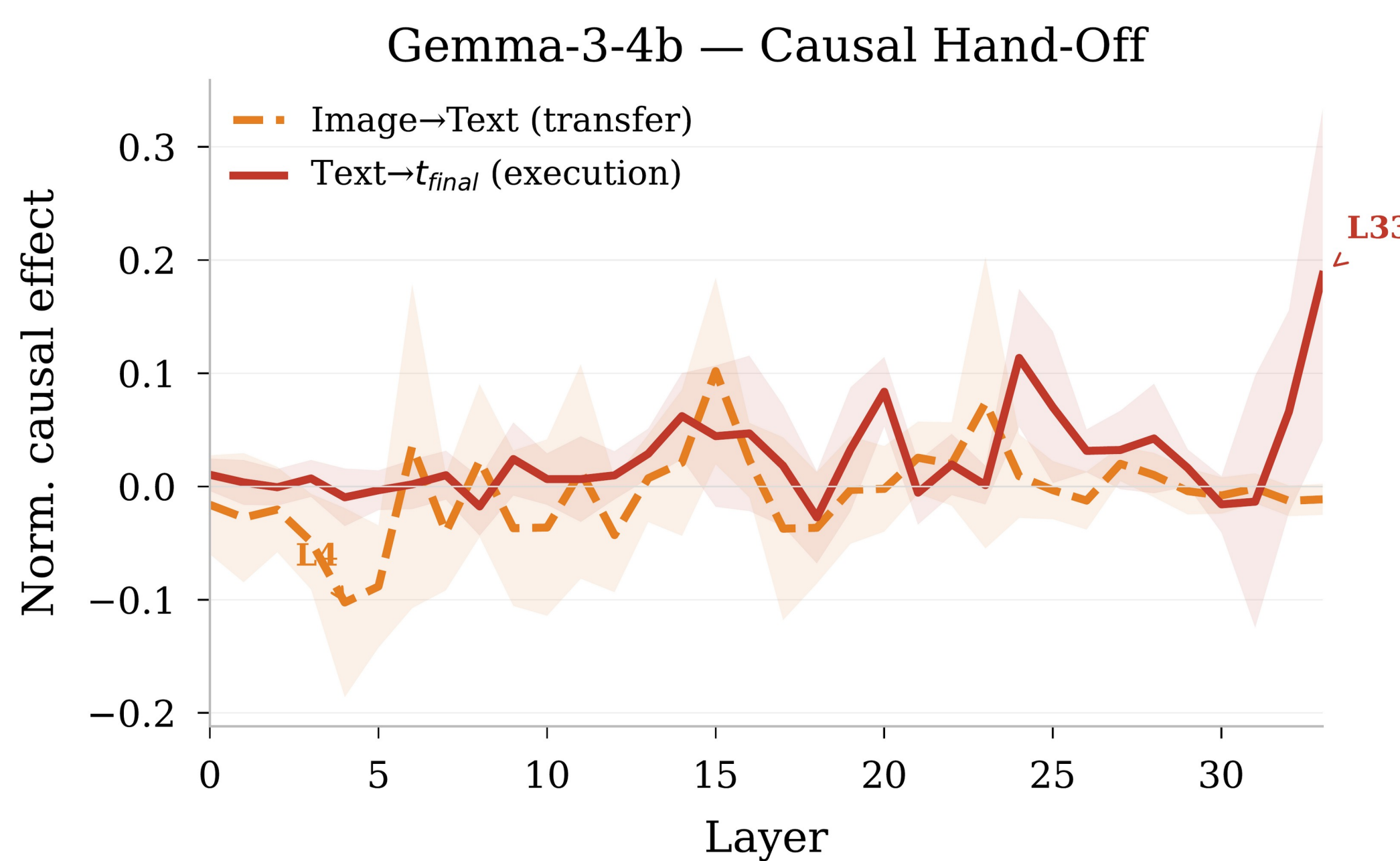
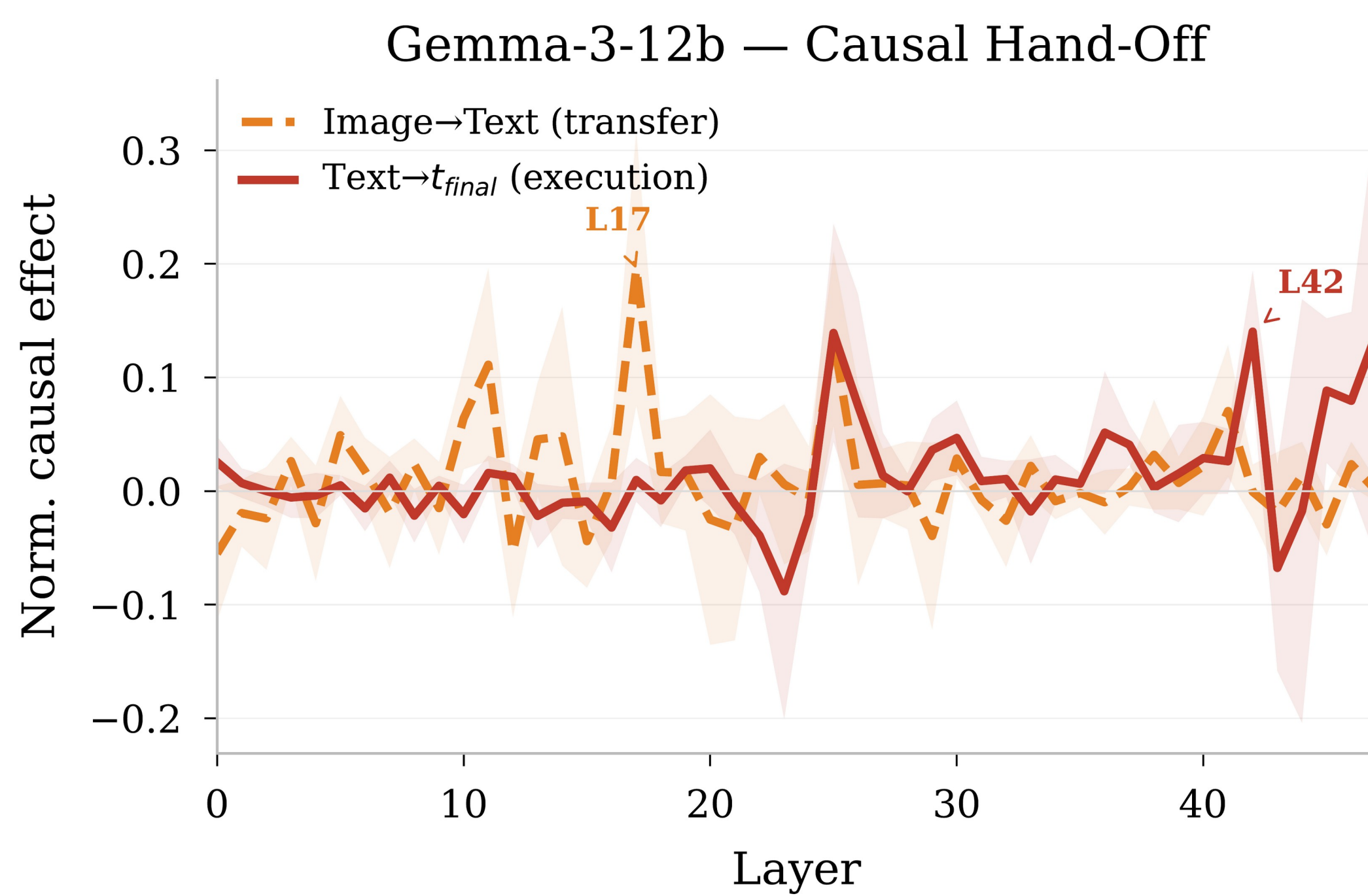


Image enters text **early** and steers the decision **late**



Late fixes cannot undo early fusion → **defer instead**

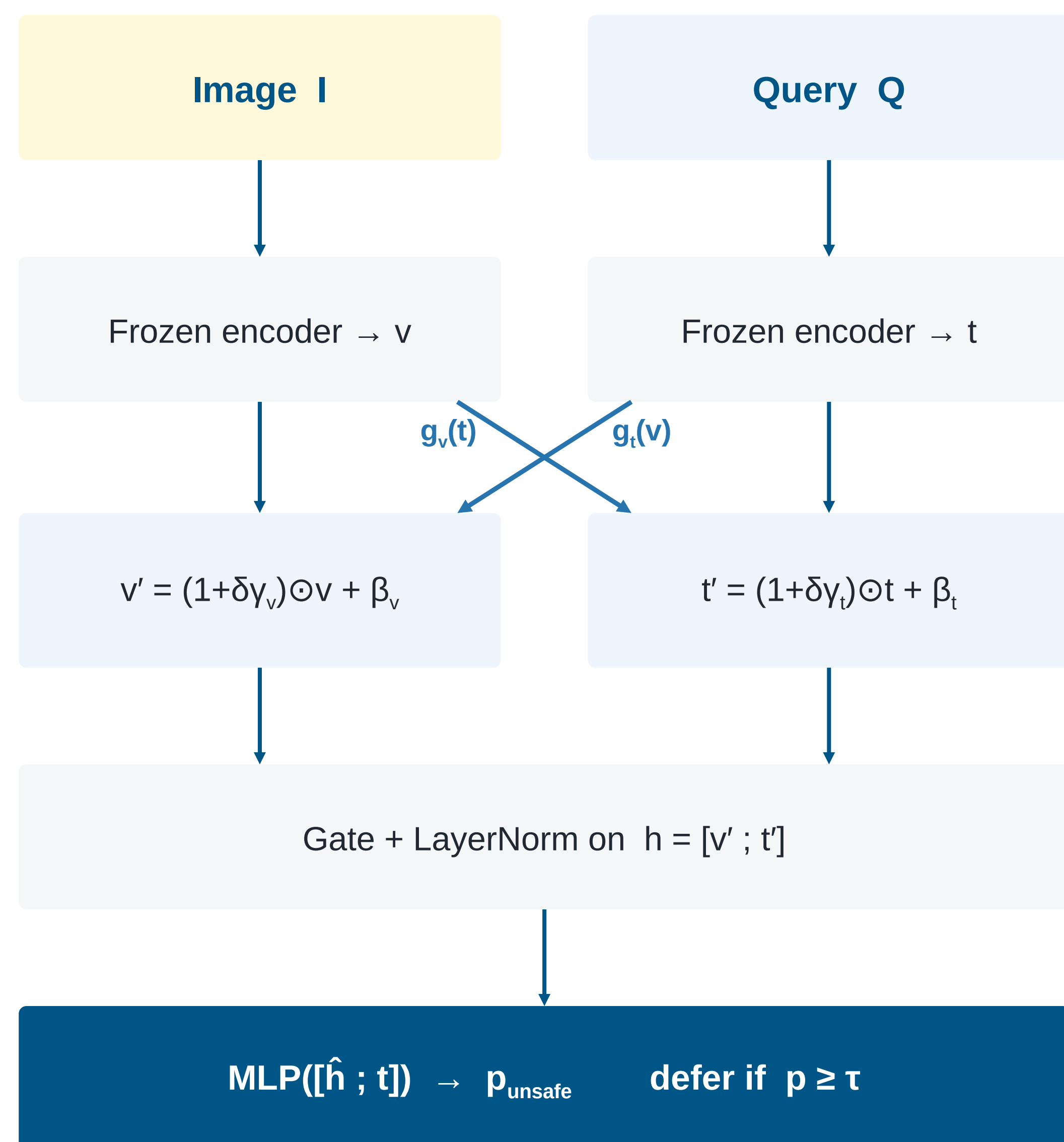
## 7 PRISM Knows When to Defer

## Flags queries that need user context **before** answering

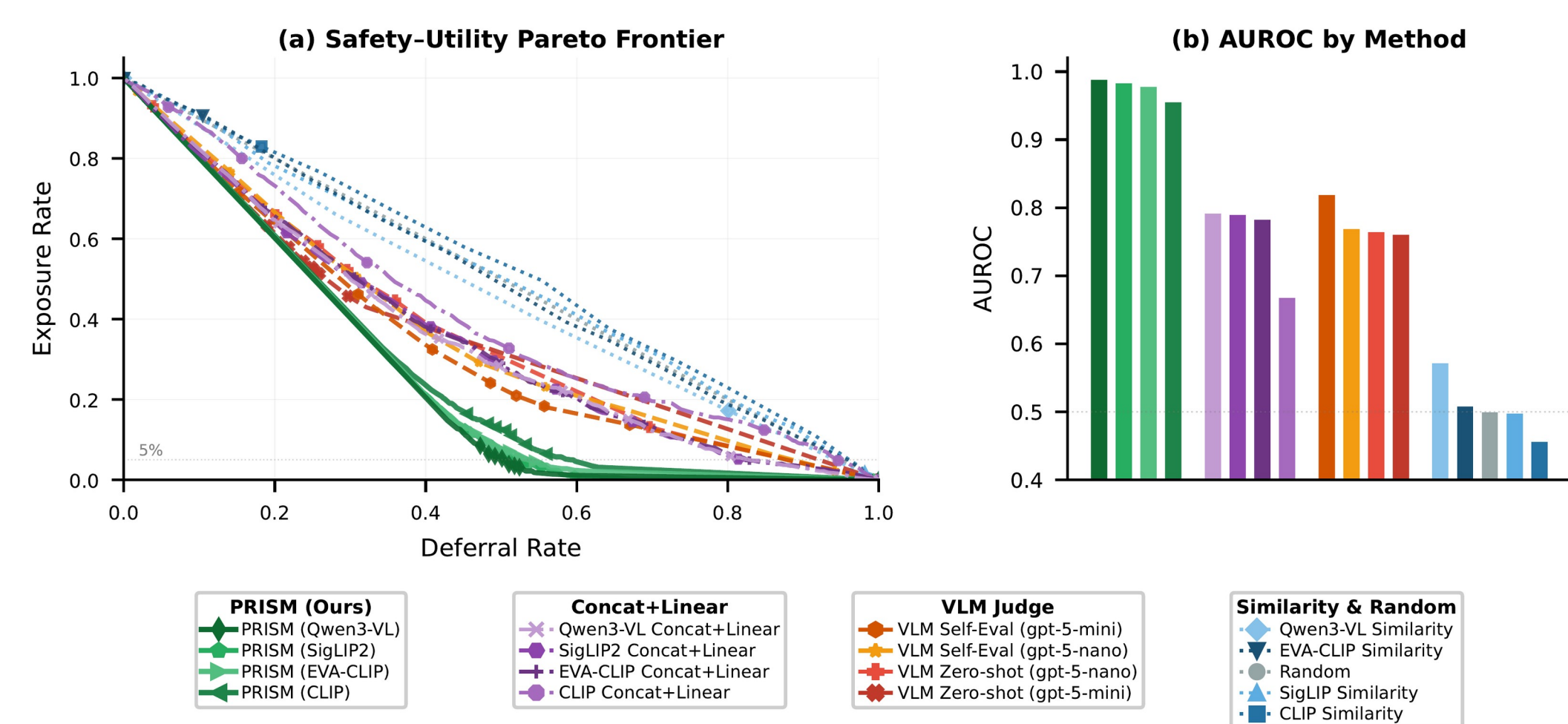
0.95–0.98  
AUROC

**10–20M**  
trainable params

## 4 encoder backbones



### Two-way modulation counters visual dominance



- Beats VLM judges (0.76–0.82) and linear probes (0.67–0.79)
- Lowest unsafe exposure at every deferral rate

[seeing-overrides-knowing.github.io](https://seeing-overrides-knowing.github.io)